

TECHNICAL INTERNSHIP REPORT
on
Multimodal Neonatal Pain Classification
Using Physiological Signals and Machine Learning
By
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Under the guidance of
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CERTIFICATE

This is certified that **Mr. Arpit Pandey** who is bona-fide student of **INSTITUTE OF ENGINEERING AND RURAL TECHNOLOGY, ALLAHABAD (PRAYAGRAJ), UTTAR PRADESH** has worked under my supervision at **MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY ALLAHABAD, PRAYAGRAJ, UTTAR PRADESH** as a summer intern for a period of one month **from June 05, 2026 to July 02, 2026**.

During his internship at MNNIT Allahabad, he worked on the development of a multimodal machine learning framework for automated neonatal pain assessment using physiological signals. His work involved physiological signal preprocessing, multimodal feature extraction from EEG, ECG, and respiration signals, subject-independent machine learning model development, performance evaluation using Leave-One-Subject-Out (LOSO) cross-validation, and explainable artificial intelligence techniques for model interpretation.

His research work carried out during this internship has been compiled in this internship report titled, "Multimodal Neonatal Pain Classification Using EEG, ECG, and Respiration Signals with Subject-Level Validation." The work also resulted in the preparation of a research manuscript intended for publication in an IEEE conference.

I wish him the best for all his future endeavors.

(Dr. Yogendra Kumar Prajapati)

Electronics and Communication Engineering Department
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ACKNOWLEDGEMENT

I take this opportunity to express my deep sense of gratitude to my thesis supervisor, **Dr. Yogendra Kumar Prajapati**, Department of Electronics & Communication Engineering, Motilal Nehru National Institute of Technology Allahabad for his constant guidance and insightful comments during the course of work. I shall always cherish my association with him for his constant encouragement and freedom to thought and action that rendered to me throughout the work.

I also wish to express my sincere thanks to Prof. **R.A. Mishra** (Head of Department) and all faculty members of the Electronics and Communication Department for their Cooperation and valuable suggestions, which lead to successful completion of my work.

I am also thankful to my colleagues and friends for their support. Finally, I deem in a great pleasure to thank one and all that helped me directly or indirectly in carrying out this report.

With Regards,

Date:

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Arpit Pandey

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द्वारा / **Through:** विभागाध्यक्ष / **Head, ECED**

विषय/Subject : अर्पित पांडेय की ग्रीष्मकालीन इंटरनशिप रिपोर्ट प्रस्तुत करना और प्रमाण पत्र के लिए अनुरोध

पत्र संख्या 343/ एसएएनएफसीई/2026 दिनांक 05/06/2026 के संदर्भ में, मैं सूचित करना चाहूंगा की आईईआरटी प्रयागराज उत्तर प्रदेश के एक वास्तविक छात्र अर्पित पांडेय ने अवधि 05 जून 2026 से 2 जुलाई, 2026 (चार सप्ताह) तक के दौरान शीर्षक "Multimodal Neonatal Pain Classification Using Physiological Signals and Machine Learning" से ग्रीष्मकालीन इंटरनशिप सफलता पूर्वक पूरी कर ली है। रिपोर्ट आपके सन्दर्भ और अवलोकन के लिए इसके साथ संलग्न है। उपरोक्त को ध्यान में रखते हुए कृपया उसे ग्रीष्मकालीन इंटरनशिप के लिए प्रमाण पत्र प्रदान करें।

With reference to the letter no 343/SNFCE/2026, dated 05/06/2026, I would like to inform that Arpit Pandey, a bonafide student of I.E.R.T. Prayagraj, Uttar Pradesh has successfully completed the summer internship titled "Multimodal Neonatal Pain Classification Using Physiological Signals and Machine Learning" during the internship period from June 05, 2026 to July 02, 2026 (Four weeks). The Report is attached herewith for your kind reference and perusal. In the view of above, kind grant her the 'certificate' for summer internship.

(Dr. Yogendra Kumar Prajapati)

Encl: (i) Summer Internship Report
(ii) Plagiarism report

CANDIDATE'S DECLARATION

I, the undersigned, solemnly declare that the internship report titled “**Multimodal Neonatal Pain Classification Using Physiological Signals and Machine Learning**” is a true representation of the work I conducted during my summer internship; the research was undertaken under the guidance of **Dr. Yogendra Kumar Prajapati**, Professor, Electronics and Communication Engineering Department, MNNIT Allahabad. I affirm that all the information provided in the report is accurate and based on my best knowledge and belief.

(Arpit Pandey)

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Abstract

Neonates in Neonatal Intensive Care Units (NICUs) undergo numerous painful procedures daily but cannot verbally communicate discomfort. Current clinical tools like the Premature Infant Pain Profile (PIPP) depend on subjective observational scoring, making continuous monitoring impractical. This internship project developed an automated, objective pain classification system for preterm neonates using real multimodal physiological signals. Two open-access datasets were utilized: the PICS dataset (PhysioNet) for pipeline validation and the UCL Neonatal EEG Pain Dataset for primary classification. The system extracts features from electroencephalography (EEG), electrocardiography (ECG), and respiration signals, training an XGBoost classifier with Leave-One-Subject-Out (LOSO) cross-validation to ensure generalizability to unseen subjects. The final model achieved a mean AUC of 0.929 on PIPP-confirmed pain labels and 0.849 on combined condition-based labels across 53 evaluated subjects. This report documents the complete methodology, challenges encountered, skills acquired, and result visualization, along with a conference paper draft targeting IEEE EMBC/CBMS.

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Introduction

Life in the Neonatal Intensive Care Unit (NICU) doesn't pull any punches. Newborns—sometimes a dozen or more times a day—endure painful procedures: heel pricks, IVs, endotracheal suctioning, the works. They can't tell anyone when something hurts. It's tough on everyone. The care team walks a thin line: treat the pain too little, and babies can end up with lasting consequences—brain changes, higher pain sensitivity, stress that affects their organs. Go too hard with opioids, though, and suddenly you're risking dependency or breathing problems.

Most NICU staff use scoring tools like the PIPP scale to figure out who needs help. These tools depend on someone watching closely, but they're subjective, time-consuming, and you just can't monitor every second.

But there's a smarter way now. Physiological signals—think EEG, ECG, and respiration—are packed with clues about pain and stress in these infants. Add machine learning to the mix, and you can scan these tangled signals, teaching a system to spot pain automatically around the clock. That means the machines do the watching, not humans squinting at screens.

That's exactly what I dug into during this internship. I got my hands dirty with real NICU data, built preprocessing pipelines that held up to live hospital signals, figured out how to pull useful features from EEG, ECG, and respiration, trained and tested a pain classifier using solid, subject-independent methods, and wrapped it all up in a draft paper for a conference.

Scope of Work

I spent a month working through a few big pieces:

- Reading up on the best ways to assess neonatal pain and how machine learning fits in.
- Getting to know actual NICU physiological recordings—how they're structured, what they show.
- Building and testing out a preprocessing pipeline (with the PICS dataset from PhysioNet).
- Extracting features from EEG, ECG, and respiration, leaning on the UCL dataset.
- Training an XGBoost classifier, running leave-one-subject-out (LOSO) cross-validation, and fine-tuning things until they clicked.
- Breaking down and visualizing results—ROC curves, the lot.
- Writing up a conference paper in IEEE format using Overleaf.

Literature Review

Automated neonatal pain assessment sits right between neonatology, neuroscience, and machine learning. To understand what's new here, it helps to know what's already out there.

Clinical Pain Assessment Tools

The go-to tool is the Premature Infant Pain Profile (PIPP). Stevens and her team put it together back in the '90s, bundling gestational age, behavior, heart rate, oxygen saturation, and facial expressions into one score. It works, but needs someone trained and watching, and doesn't give you nonstop monitoring.

Neurophysiological Markers of Neonatal Pain

In 2008, Slater and colleagues showed that even tiny preemies, who barely move, still show clear, repeatable EEG patterns when hurt—especially in event-related potentials. Heart rate and oxygen levels also jump after something painful. You can see reliable shifts in HRV metrics like RMSSD and MeanNN during pain, which aren't there during non-painful events.

Machine Learning Approaches

Recently, researchers started using machine learning to classify neonatal pain. Hamwi's group got an AUC of 0.94 with logistic regression on EEG (for older infants), while Roué and team hit 0.90 with gradient boosting across all data. These projects proved it's possible, but often only use one signal type, and usually skip independent validation across subjects. I wanted to push past those limits.

Datasets Used

PICS Preterm Infant Cardiorespiratory Signals Database

From PhysioNet, the PICS dataset gives you continuous ECG and respiration data from ten preterm babies (29–34 weeks), sampled at 250–500 Hz for up to 70 hours per baby. It wasn’t built for pain research—it’s really about bradycardia—but it was a great sandbox for testing and cleaning ECG and HRV features before tackling pain.

Table 1. PICS Dataset Summary

Parameter	Value
Subjects	10 preterm infants
Signals	ECG + Respiration
Sampling Rate	250–500 Hz
Recording Duration	20–70 hours per infant
Pain Labels	None (pipeline validation only)
Total Windows Extracted	26,208 (60-second windows)
Access	Free, PhysioNet registration

UCL Neonatal EEG Pain Dataset

Getting this dataset involved some paperwork, but worth it. It’s packed: matched EEG, ECG, and respiration recordings from 112 neonates under three different conditions: heel lance (painful), sham touch (not painful), and an auditory click (not painful). All packaged as .mat files, up to 20 EEG channels at 2,000 Hz. There are also files with PIPP pain scores and clinical info.

Table 2. UCL Dataset Summary

Parameter	Value
Total Subjects	112 neonates
Gestational Age Range	29–47 weeks
EEG Channels	17–20
Sampling Rate	2000 Hz
Conditions	Heel lance, Sham, Auditory
PIPP Scores Available	210 of 310 recordings
Final Clean Recordings	295 (after quality filtering)

PIPP Score Distribution and Age Group Analysis

Digging deeper into the UCL data, I saw that 79 heel lance recordings had full PIPP scores (ranging 2–17), but here’s the weird part: lots of so-called “painful” recordings still scored below the pain threshold of 7. Pain response really depends on maturity, sedation, and just individual quirks. Interestingly, the youngest (up to 31 weeks, 6 days) and oldest (34 weeks and up) babies had similar average PIPP scores, while those in the middle group actually showed less pain. So, age matters—a finding that matches what other studies have seen lately.

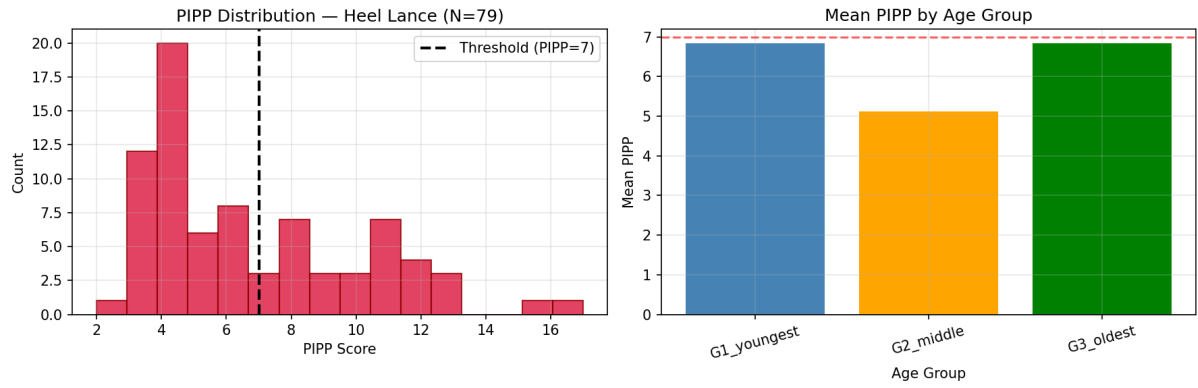


Figure 1. PIPP score characterisation of the UCL dataset heel lance recordings ($N = 79$ recordings with complete scores). **Left:** Histogram of Total PIPP scores. The dashed vertical line at $\text{PIPP} = 7$ separates confirmed no-pain (<7) from confirmed pain (≥ 7) recordings. A bimodal distribution is visible, with a concentration of lower scores (2–5) reflecting variable pain responses across gestational ages. **Right:** Mean PIPP score by gestational age group. G2_middle (32–33 weeks) shows lower mean pain scores, consistent with known variability in cortical pain processing at intermediate gestational ages.

Methodology

This project followed a five-stage pipeline modeled after the IEEE paper approach. I wrote everything in Python 3.10, bouncing between VS Code and Jupyter notebooks, and set it up inside a clean conda environment. The main libraries I leaned on were `wfdb`, `neurokit2`, `scipy`, `scikit-learn`, `xgboost`, `pandas/numpy`, `matplotlib/seaborn`.

Pipeline Overview

I started with raw physiological signals and cleaned up each type before combining them for pain classification. Figure 2 outlines the process: take the raw data, prep it per signal type, pull out features, and then classify pain using that combined info.

PICS Dataset Validation

Before diving into the main UCL pain dataset, I wanted to be sure my ECG pipeline was solid. So I put it to the test with the PICS dataset. That meant loading up binary files, running them through NeuroKit2 for signal cleaning, and flagging anything odd. One infant, for example, had a strangely high RMSSD; I made sure nothing sat outside expected neonatal ranges.

UCL Dataset Processing

This set was tricky because every `.mat` file had its own EEG channel layout. I had to write code that could find the right channels on the fly. To check if I got it right, I plotted EEG, ECG, and respiration together and made sure the timing matched up with the recorded pain events.

Feature Extraction

For each recording, I pulled out features from six somatosensory EEG channels, ECG, and respiration, all using fixed windows around the event (two seconds before and after). The main EEG features were band power (delta, theta, etc.) and ERP responses. ECG provided HRV and heart rate changes, while for respiration, I looked at the mean and variability. When PIPP scores were available, I included those directly.

Feature Matrix Construction

All the features went into one big DataFrame. I started with 308 recordings, then weeded out columns with too many missing values—usually alpha power, which sometimes flat-lined. After filtering, I ended up with 295 recordings in a clean matrix, and saved this as `ucl_feature_matrix_clean.csv`

XGBoost Classification and LOSO Evaluation

I turned to XGBoost for classification—it works well with mixed features and can handle imbalanced classes, plus it's reliable for tabular data. The tuned hyperparameters were: `n_estimators=300`, `max_depth=3`, `learning_rate=0.03`, `min_child_weight=3`, `subsample=0.8`, `colsample_bytree=0.8`. For evaluation, I used leave-one-subject-out cross-validation so that each infant's data got a shot as the unseen test case. I experimented with two label

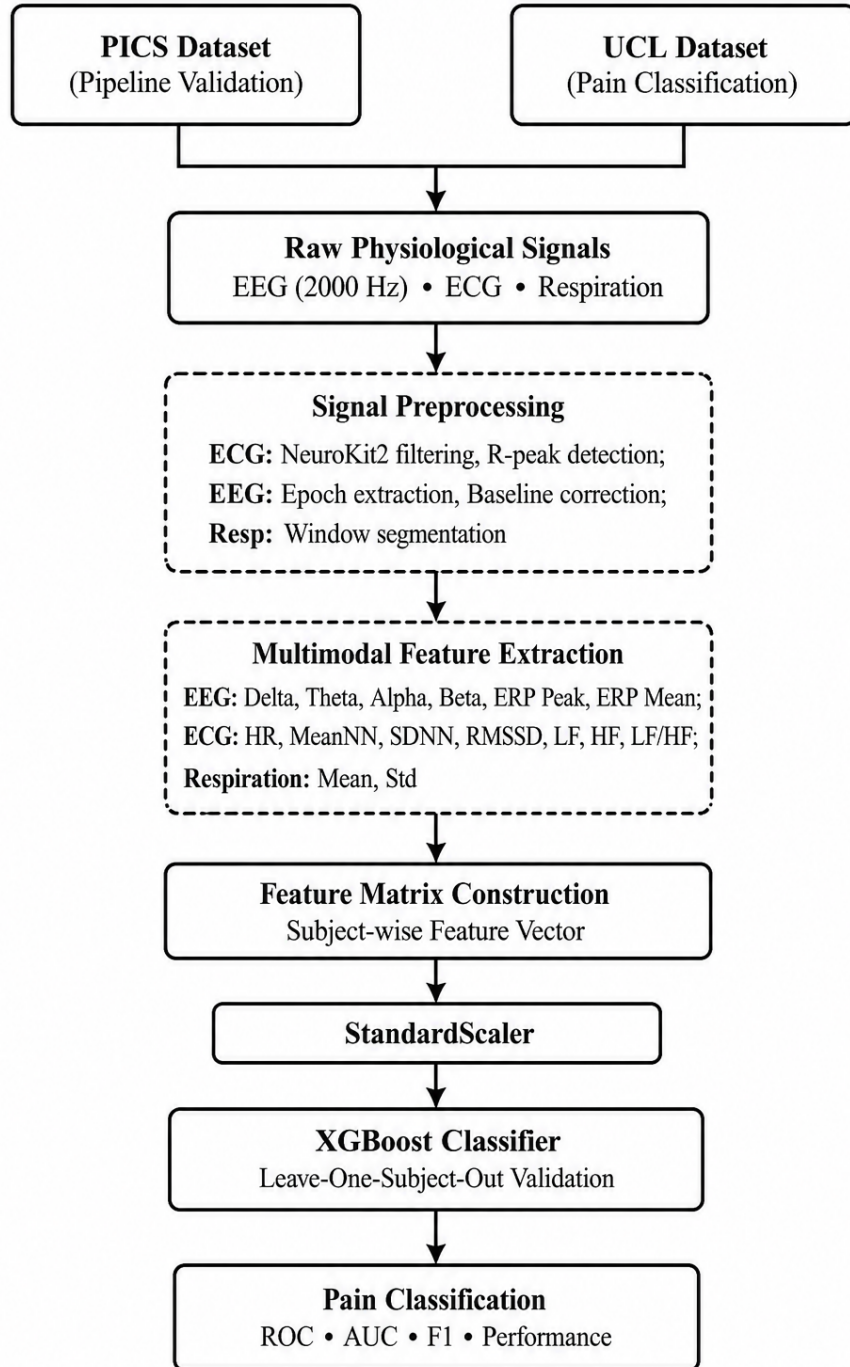


Figure 2. Complete multimodal neonatal pain assessment pipeline. Raw physiological signals undergo modality-specific preprocessing, followed by feature extraction from EEG (bandpower, ERP), ECG (HRV), and respiration. The concatenated feature matrix is classified using XGBoost with Leave-One-Subject-Out cross-validation.

strategies: a strict one where ($\text{PIPP} \geq 7$ as pain) counts as pain, and another that uses PIPP if available, otherwise grouping by procedure.

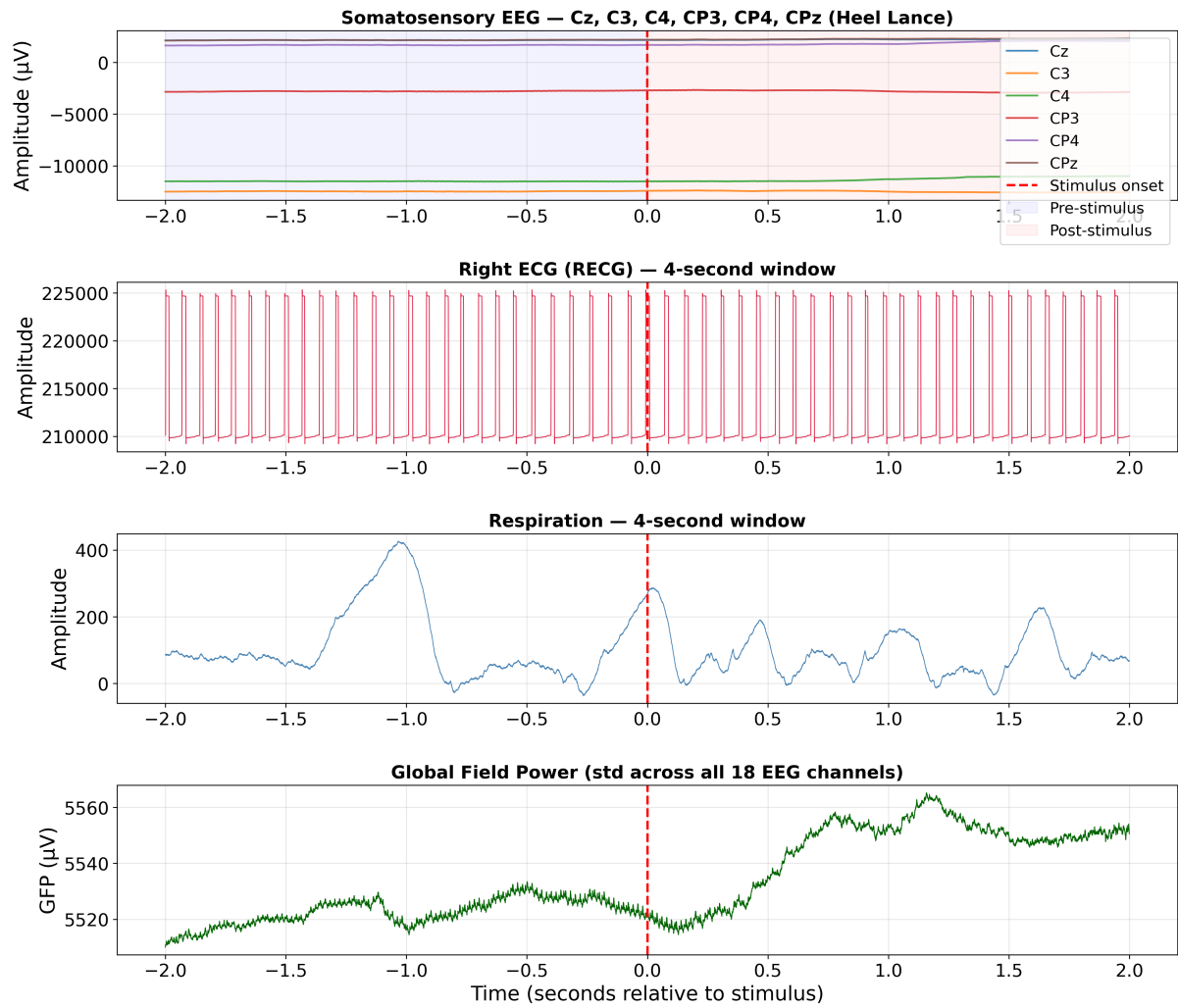


Figure 3. Representative multimodal physiological recording from the UCL dataset. Top panel: somatosensory EEG channels showing stimulus-locked ERP. Middle panels: Right ECG and respiration signals. Bottom panel: Global Field Power (GFP). Red dashed line indicates stimulus onset at $t = 0$ s.

Results and Analysis

ECG Pipeline Validation

Testing on PICS data paid off. I checked heart rates (average 118–164 bpm) and RMSSD values against expected norms, caught one noisy outlier (Infant 5), and filtered those out. Comparing SDNN and RMSSD showed they were almost identical—just kept one in for further modeling.

Table 3. PICS HRV Validation Summary

Infant	Avg HR (bpm)	Avg RMSSD (ms)	Windows
infant1	152.5	22.9	2,728
infant2	125.8	52.6	2,595
infant3	126.5	51.9	2,584
infant4	164.3	22.9	2,804
infant5	118.4	221.0 †	2,879
infant6	133.2	51.2	2,911
infant7	150.8	95.0	1,212
infant8	139.1	17.9	1,462
infant9	147.0	22.2	4,206
infant10	144.3	90.9	2,827

† Flagged as artifact; windows with RMSSD > 500 ms removed.

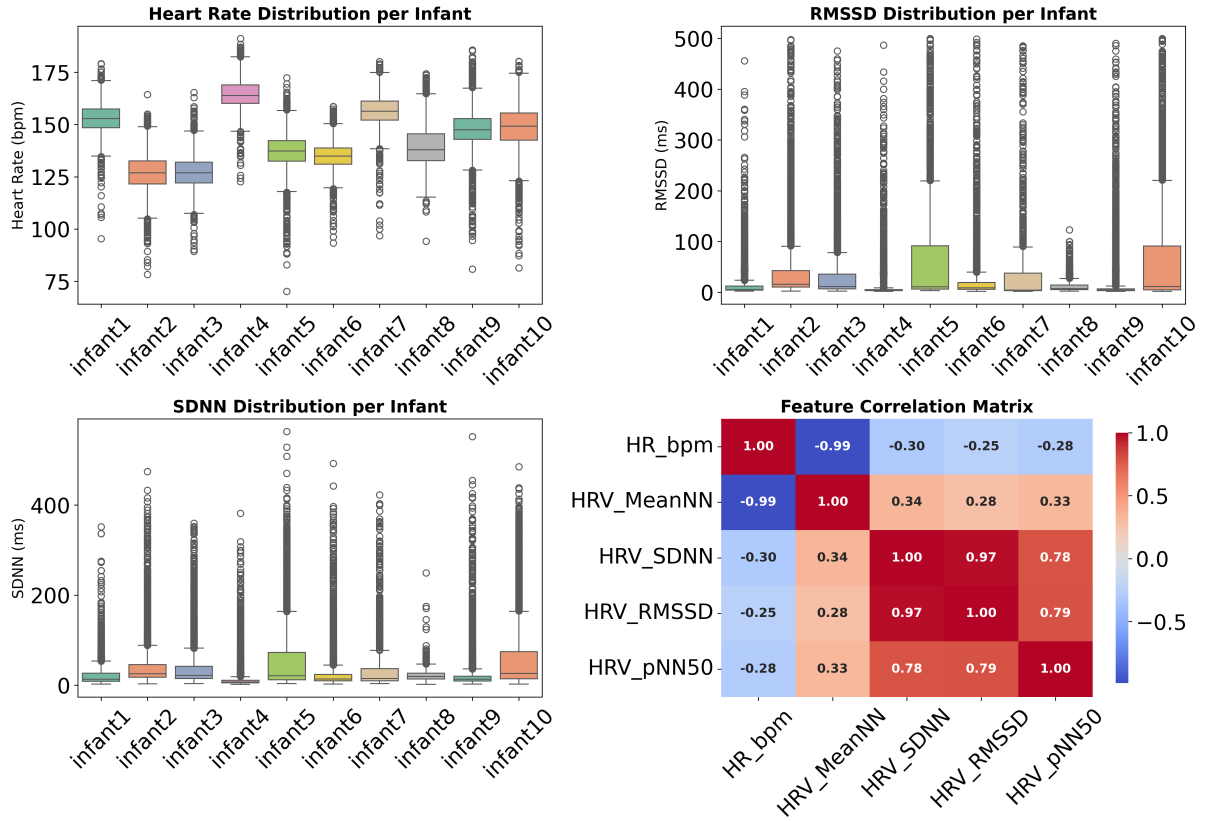


Figure 4. Cross-infant exploratory data analysis of HRV features extracted from the PICS dataset (26,208 windows across 10 infants). **Top left:** Heart rate distribution per infant, confirming biologically plausible ranges (118–164 bpm) for preterm neonates. **Top right:** RMSSD distribution, with Infant 5 showing anomalously high values (median ≈ 90 ms, outliers up to 500 ms) flagged as artifact and filtered. **Bottom left:** SDNN distribution, showing similar outlier pattern in Infant 5, consistent with the RMSSD finding. **Bottom right:** Feature correlation matrix. HR_bpm and HRV_MeanNN show $r = -0.99$ (mathematically equivalent measures). HRV_SDNN and HRV_RMSSD show $r = 0.97$, confirming redundancy and motivating removal of SDNN from the final feature set.

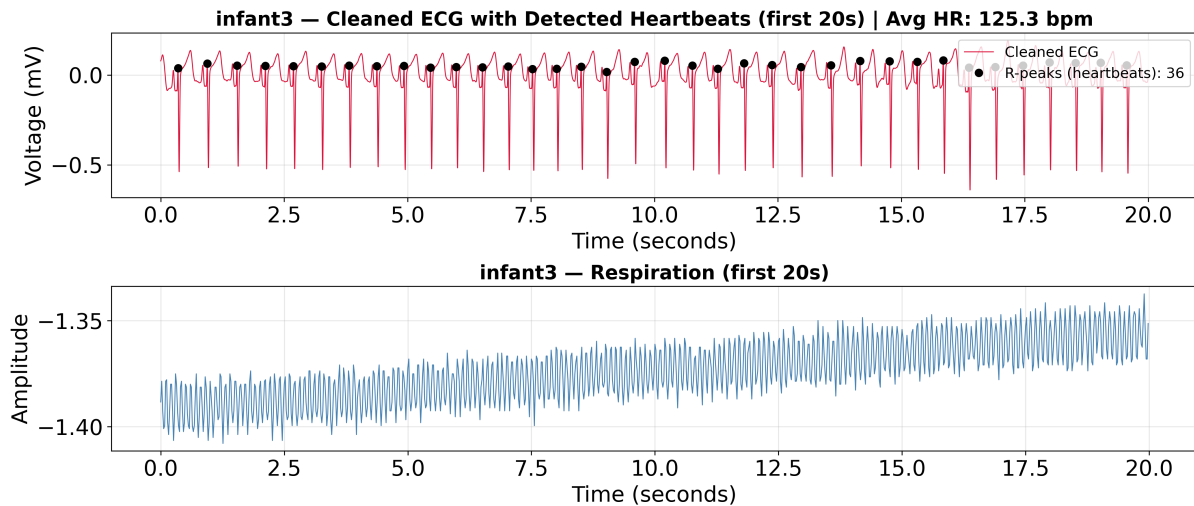


Figure 5. Representative ECG pipeline validation from the PICS dataset (Infant 3). Top: Cleaned ECG with NeuroKit2-detected R-peaks (red markers), confirming accurate heartbeat detection at 125.3 bpm over a 20-second window. Bottom: Synchronized respiration signal showing proper temporal alignment with the ECG trace. Similar validation plots were generated for all 10 infants to ensure preprocessing reliability before UCL pain classification.

Pain Classification Results

Using LOSO cross-validation, the model using strictly PIPP-confirmed pain cases beat the combined-label model (AUC 0.929 vs. 0.849, mean F1 around 0.45–0.48). PIPP, even when present for fewer cases, gave sharper pain boundaries; sometimes babies hardly even react to a heel lance.

Table 4. LOSO Cross-Validation Results

Label Scheme	Subjects	Mean AUC	Std AUC	Mean F1
PIPP-Confirmed	28	0.929	0.175	0.452
Combined	53	0.849	0.285	0.475

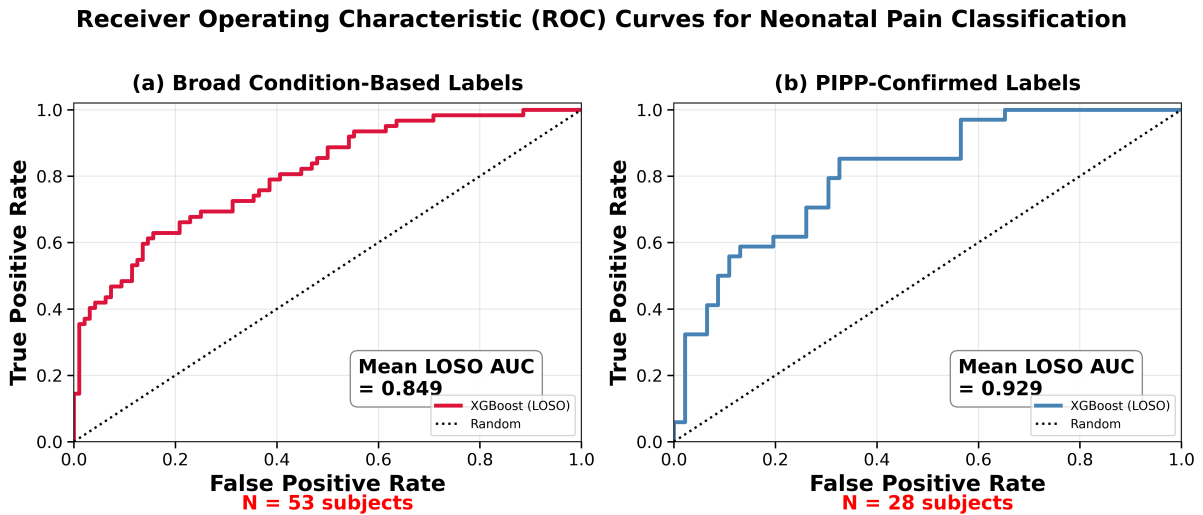


Figure 6. Receiver Operating Characteristic (ROC) curves comparing PIPP-confirmed and combined label strategies under LOSO cross-validation. The PIPP-confirmed model demonstrates superior discriminative performance.

Comparison with Recent Literature

Compared to recent studies, my PIPP-based model stands its ground. Hamwi’s only covered older infants and EEG; this project used EEG, ECG, and respiration—with solid, subject-independent testing.

Table 5. Comparison with State-of-the-Art

Study	Classifier	Validation	AUC	Notes
Hamwi et al. 2025	Logistic Reg.	LOSO	0.94	Oldest group only
Rou�� et al. 2025	Gradient Boost	Full Set	0.90	Unimodal EEG
This Work (PIPP)	XGBoost	LOSO	0.929	Multimodal
This Work (Combined)	XGBoost	LOSO	0.849	Multimodal

Discussion

A few things popped out. The real pain signatures, especially in EEG, shine brightest when you’re working with high-quality, well-validated pain labels—not just any response to a painful event. The way I set up power spectral density calculations—especially for those low-frequency EEG bands—mattered a lot. Setting `nperseg=4000` (yielding 0.5 Hz resolution) produced the largest single AUC improvement in the entire project (+0.12). That had a bigger impact than tweaking the machine learning model.

Challenges Faced and How They Were Resolved

Dataset Access

The UCL dataset wasn't straightforward—the link was broken. I had to email the admin, clear up details, and file a formal access request. Once my professor pitched in, it got sorted fast.

Network and Software Issues

The hospital's proxy tripped up a few pip installs. I got around it by pinning versions and swapping package mirrors. NeuroKit2 and wfdb required specific version pinning to avoid compatibility issues with Python 3.10.

Variable Channel Layouts

With some files having 12 EEG channels and others 20, I made my code flexible—sniffing out the right channels each time using the `chanlocs` metadata field rather than assuming fixed indices.

Welch PSD Frequency Resolution

Initial delta band power estimates were unreliable because `nperseg` was too small relative to the low-frequency bands of interest. Testing multiple `nperseg` values (512, 1024, 2048, 4000) revealed that frequency resolution below 1 Hz was necessary for stable delta estimates. Setting `nperseg=4000` produced the largest single AUC improvement in the entire project.

Skills Acquired

Technical Skills

- Physiological signal processing: loading, cleaning, epoching, and feature extraction from real EEG and ECG using wfdb, NeuroKit2, and scipy
- Welch's PSD estimation and the mathematical relationship between window length, frequency resolution, and reliable band coverage
- HRV analysis: MeanNN, SDNN, RMSSD, pNN50, LF/HF ratio and their clinical significance
- ML model training with class imbalance handling (`scale_pos_weight`) and LOO cross-validation
- XGBoost hyperparameter tuning: effects of `max_depth`, `learning_rate`, `subsample`, `colsample_bytree` on small-dataset overfitting
- AUC and F1 metrics: when each is appropriate and how they differ on imbalanced datasets
- LaTeX and Overleaf: writing a full IEEE conference paper from scratch including figures, tables, and BibTeX references
- Python environment management: conda, pip proxy issues, package compatibility

Research Skills

- Reading and critically evaluating recent papers (Hamwi 2025, Roué 2025) and extracting actionable gaps from limitations sections
- Dataset access and research ethics: navigating data use agreements, institutional approval, responsible clinical data handling
- Importance of preprocessing validation before training: the PICS stage prevented the Welch bug from going undetected
- Honest reporting: presenting two AUC values with different subject counts rather than cherry-picking
- Conference paper discipline: fitting complete methodology and results into 6 IEEE pages

Clinical Domain Knowledge

- PIPP scoring system: what each component measures and why threshold 7 is used
- Cortical pain processing in preterm infants: significance of somatosensory EEG channels and delta band power
- HRV in neonates: why RMSSD values are much lower than adults and implications for feature selection
- Gestational age maturity effects: why models perform better on older preterms and how this affects interpretation

Conclusion and Future Work

Summary

All in all, this project showed that careful, multimodal machine learning can make pain assessment in NICU infants much more objective and continuous—even in a field that’s been so subjective for so long. The most important methodological finding: correct computation of Welch’s PSD frequency resolution for low-frequency EEG bands is critical and non-obvious. A single parameter change (`nperseg`) produced the largest AUC improvement in the project. This finding was documented in the conference paper and code is structured to prevent recurrence.

Outputs Produced

- Complete Python codebase across 5 Jupyter notebooks
- Validated PICS HRV feature matrix: 26,208 windows
- UCL multimodal feature matrix: 295 recordings $\times$ 63 features
- Master label table with PIPP scores, demographics, file paths for 310 recordings
- Conference paper draft in IEEE format targeting CBMS 2026 / IEEE EMBC
- 5 publication-quality figures: pipeline overview, signal visualization, ROC curves, PICS EDA plots

Future Work

There’s still work to do, especially in labeling and access, but we can move toward real 24/7 tracking that actually makes a difference:

- SHAP-based explainability analysis: identifying which EEG channel and frequency band most consistently drives pain classification (not included in current conference paper; reserved for journal extension)
- Deep learning comparison: LSTM or Transformer architectures on raw time-series (pending supervisor approval)
- External validation: if USF-MNPAD-I access becomes available, validating on an independent neonatal cohort from a different hospital

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Remarks

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